Lecture 20 — Transforming space

Wed Oct 14 and Fri Oct 16

Last updated: October 19, 2026. Taught across Wednesday, October 14 and Friday, October 16.

Boyd video (optional): Lecture 19 — matrix–vector / transform examples · Lecture 23 — vectors as linear functions · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 7.1–7.2 and Lachniet Ch. 2. Related objectives: Lectures. Schedule.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Predict the effect of scaling, reflection, shear, and rotation matrices.
2. Apply matrices to many 2D or 3D points at once.
3. Plot and compare original and transformed objects.
4. Adapt a short program that implements a transformation.
5. Interpret a matrix as a coordinate or physical-space transformation.

Warm-up

Predict, without multiplying: where does $A = \mathrm{diag}(2, 1/2)$ send the point $(1, 1)$? Sketch the original and image roughly.

Fill in

Many geometric maps on $\mathbb{R}^2$ (or $\mathbb{R}^3$) are matrix-vector products $y = Ax$.

Finding the matrix. The jth column of A is what the map does to _____. So apply the geometry to e_1, e_2 (and e_3 in 3D) to read off columns.

Common 2×2 examples:

- **Scaling / dilation:** $A = \mathrm{diag}(s_1, s_2)$ stretches the axes by s_1, s_2.
- **Rotation** counterclockwise by θ:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

- **Reflection** through a line through the origin; **shear** (e.g. $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$).

Many points at once. If columns of X are points, then AX applies the same map to every column.

Change of coordinates (3D idea). Body-fixed vs earth-fixed frames: $x_{\text{earth}} = p + Qx_{\text{body}}$, where columns of Q are the body axes expressed in earth coordinates.

Worked in class

W1. Build from e_1, e_2. Reflection through the x_1-axis sends $(1, 0) \mapsto (1, 0)$ and $(0, 1) \mapsto (0, -1)$. Write A. Check $A(2, 3)$.

W2. Rotation. For $\theta = 90^\circ = \pi/2$, write the rotation matrix and compute $A(1, 0)$ and $A(0, 1)$. Does that match "90° counterclockwise"?

W3. Many points. Let $X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (three corners of a right triangle). With $A = \text{diag}(2, 1)$, compute AX and describe the image.

W4 (time permitting). Shear $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Where do $(1, 0)$ and $(0, 1)$ go? Sketch the unit square's image.

Your turn

Y1. Reflection through the x_2-axis: write A by tracking e_1 and e_2, then compute $A(3, -1)$.

Y2. Rotation by 180°: write A and apply it to $(2, 1)$.

Y3. True or false: every 2×2 matrix represents a rotation of the plane.

Answers (Your turn)

Y1. $e_1 \mapsto (-1, 0)$, $e_2 \mapsto (0, 1)$, so $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. $A(3, -1) = (-3, -1)$.

Y2. $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$. $A(2, 1) = (-2, -1)$.

Y3. **False.** Scaling, shear, projections, and many other maps are also 2×2 matrices; rotations are a special family.